

S-Plane Master Test Catalogue — Addendum

Coverage-gap closure (development) · 2026-09-20 · Companion to [ORAN_SPlane_Master_Test_Catalogue.pdf](#) · Frozen rule `decision_rule.py sha256 c362e11...` (unchanged, hash-verified)

What this addendum is. A controlled test of three catalogue items previously marked “not run,” plus a development detector set that closes them. It is **not** a live on-wire result and **not** a new frozen artefact. No live execution environment was available this session (device VM unprivileged — network-namespace creation denied, no ptp4l/scapy; cloud shell blocked), so each item was tested by feeding the **frozen rule** crafted deep-CSV inputs that differ from a G.8275.1-conformant baseline by exactly one injected signature — the same detection-logic method as `conformance_neg.py`.

Result 1 — the three gaps are real (frozen rule, hash-verified)

Item	Class (standard)	Crafted input (only delta from baseline)	Truth	Frozen verdict
—	Conformant control	<code>synth_base</code>	BENIGN	BENIGN — control passes
1	Packet removal / selective interception (O-RAN WG11 11.1.5.3.1)	70% of Sync/Follow_Up/Delay_Response dropped	AT TA C K	MISS BENIGN
2	Malformed / fuzzed frames (O-RAN WG11 24.2.1.2)	200 GM frames: version≠2, length<min, control>5	AT TA C K	MISS BENIGN
3	Whole-second field abuse (IEEE 1588-2019 §7.2.4)	leap61=1, currentUtcOffset 37 → 0, traceability stripped	AT TA C K	MISS BENIGN

Control passing while all three attacks pass as BENIGN isolates the blindness to the injected class: the frozen rule inspects none of observed-vs-declared rate, frame legality (version/length/control), or timescale metadata (leap/UTC/traceability).

Result 2 — `decision_rule_v2` closes them **UNFROZEN / development**

Three standards-grounded detectors, each fixed by a standard or **self-referential** (compared to what the sender itself declares — no constant fitted to the crafted inputs). Runs the frozen rule first and **never turns a frozen ATTACK into BENIGN**; it only escalates BENIGN/UNKNOWN when a new detector fires.

				Real - data FP
Item	Detector	Basis	v2 verdict	
1	D1 rate starvation	Provisioned source below 0.5× the rate <i>it itself declares</i> (logSync/logAnnounce)	ATTACK &checkmark ark;	quiet
2	D2 malformed	IEEE 1588-2019 field legality: versionPTP=2, control 0–5, legal messageType, length ≥ per-type min	ATTACK &checkmark ark;	quiet
3	D3 whole-second	leap w/o provisioned window; UTCoffset non-constant or ≠true while valid; clockClass≤6 yet traceability deasserted	ATTACK &checkmark ark;	quiet
synth_base (conformant control under full v2)			BENIGN &checkmark ark; (unchanged)	—

Real-data false-positive check: D1/D2/D3 run directly on both real healthy hardware captures (netem_baseline, netem_pdv) stay **quiet on both**. Additive-only property also verified: on those captures the frozen base returns ATTACK (domain-0, out-of-profile by design) and v2 returns the **same** ATTACK — never softened.

Honest constraints carried forward

- **v2 is not frozen and not campaign-validated.** Putting D1–D3 into any headline metric requires re-freeze + a fresh randomized campaign on the live testbed.
- **No live on-wire validation this session** (no privileged execution environment).
- netem_baseline/netem_pdv are **domain-0** real captures — valid as new-detector FP controls, **not** as G.8275.1 benign-classification controls (the frozen base flags them A5 on domain, by design).
- Hardware-only items remain isolated and out of scope: GNSS spoof/jam, holdover, SyncE/EEC, oscillator drift, delay-attack measurement.

1588), synth_*.deep.csv, decision_rule_v2.py, decision_rule_FROZEN_reference.py. Every verdict reproducible from these files.